REEVALUATING AGRICULTURAL PRODUCTIVITY GAPS WITH LONGITUDINAL MICRODATA

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Abstract
Recent research has pointed to large gaps in labor productivity between the agricultural and non-agricultural sectors in low-income countries, as well as between workers in rural and urban areas. Most estimates are based on national accounts or repeated cross-sections of microsurvey data, and as a result typically struggle to account for individual selection between sectors. This paper uses long-run individual-level panel data from two low-income countries (Indonesia and Kenya) to explore these gaps. Accounting for individual fixed effects leads to much smaller estimated productivity gains from moving into the non-agricultural sector (or urban areas), reducing estimated gaps by roughly 67%–92%. Furthermore, gaps do not emerge up to 5 years after a move between sectors. We evaluate whether these findings imply a re-assessment of the conventional wisdom regarding sectoral gaps, discuss how to reconcile them with existing cross-sectional estimates, and consider implications for the desirability of sectoral reallocation of labor. (JEL: J43, O13, O15, R23)

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Teaching Slides
A set of Teaching Slides to accompany this article are available online as Supplementary Data.

1. Introduction

The shift out of agriculture and into other more "modern" sectors (e.g., manufacturing) has long been viewed as central to economic development. This structural transformation was a focus of influential early scholarship (Rosenstein-Rodan 1943; Lewis 1955; Rostow 1960; Pack 1972; Kuznets 1973; Johnston and Kilby 1978; Schultz 1988), with the issue even stretching back to Soviet debates over whether to "squeeze" farmer surplus to hasten industrialization (Preobrazhensky 1921). A more recent macroeconomic empirical literature has documented that the share of labor in the agricultural sector correlates strongly with levels of per capita income, and yet agricultural workers are many times more productive in rich countries. This creates a double disadvantage for low-income countries: Agricultural work tends to be far less productive there, yet their workforce is concentrated in this sector.[1]

Whether or not these productivity gaps across sectors reflect labor market frictions that can be remedied by policy depends on the extent to which they can reasonably be viewed as *causal impacts* rather than mainly reflecting *worker selection* or sorting. By a causal impact of sector, we mean that a given worker employed in the non-agricultural (or urban) sector is more productive than the same worker employed in the agricultural (rural) sector. In contrast, selection implies that observed differences are driven by the fact that workers of varying ability and skill are concentrated in certain sectors and not by inherent differences in sectoral productivity.

Gollin, Lagakos, and Waugh (2014, henceforth GLW) and Young (2013) explore this identification issue by applying different methods to adjust for differences in observable characteristics, particularly education. GLW show that accounting for differences in hours worked and average worker schooling attainment across sectors reduces the average agricultural productivity gap by a third, from roughly 3 to 2. GLW remain agnostic regarding the causal interpretation of the large agricultural productivity gaps that they estimate. Nonetheless, if individual schooling largely addresses selection biases by capturing the most important dimensions of worker skill, GLW's estimates would imply that the causal impact of moving workers out of the agricultural sector in low-income countries would be large.

Young (2013) examines the related question of urban-rural differences in consumption (proxied by measures of household asset ownership, education, and child health), rather than productivity, and similarly finds large cross-sectional gaps.[2] Using

1. See Online Appendix Figures A.1 and A.2, respectively, for an illustration of these differences using cross-country data.

2. Although Young (2013) primarily focuses on urban–rural gaps, he uses data on non-agricultural versus agriculture differences when urban–rural data are missing; GLW similarly use urban–rural data when lacking data on agriculture.

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retrospective information on individual birth district available in the Demographic and Health Surveys, Young shows that rural-born individuals with more years of schooling than average in their sector are more likely to move to urban areas, whereas urban-born individuals with less schooling tend to move to rural areas. Diverging from GLW, Young argues that a model of sorting on education and relatively high demand for skill in urban areas can fully explain urban–rural consumption gaps.[3]

We examine the nature of sectoral productivity gaps using long-term individual-level longitudinal (panel) data on worker productivity. We focus on two country cases—Indonesia and Kenya—for which there exist long-term panel micro datasets with relatively large sample sizes, rich measures of earnings in both the formal and informal sectors, and high rates of respondent tracking over time.[4] We examine whether workers who changed sector correspondingly experienced the large increase in earnings suggested by existing estimates, and find that the raw mean differences in productivity can be reduced by approximately 67%–92% with the inclusion of individual fixed effects, which capture all time-invariant dimensions of worker heterogeneity.

For both countries, we start by characterizing the nature of selective migration between non-agricultural versus agricultural economic sectors, and between urban versus rural residence. We show that individuals born in rural areas who attain more schooling are significantly more likely to migrate to urban areas and are also more likely to hold non-agricultural employment, whereas those born in urban areas with less schooling are more likely to move to rural areas and into agriculture. Exploiting the unusual richness of our data, in particular, the existence of measures of cognitive ability (a Raven's Progressive Matrices score), we show that higher ability individuals are more likely to move into urban and non-agricultural sectors, even conditional on educational attainment. This is an indication that conditioning on completed schooling is insufficient to fully capture differences in average worker skill levels across sectors.

We first estimate sectoral productivity differences by treating our data as a series of repeated cross-sections, following existing work. This approach implies substantial productivity gaps between the agricultural and non-agricultural sectors of roughly 70 log points, echoing previous work. Though these are somewhat smaller than GLW's main estimates of approximately 140 log points (reproduced in the left panel of Figure 1), recall that GLW's estimates using household survey data (like ours) also tend to be smaller. Conditioning on individual demographic characteristics (age and gender) as well as hours worked and educational attainment narrows the gap, but it remains large at 38–47 log points. In our main contribution, we include individual fixed effects, and this reduces the agricultural productivity gap in earnings to 24 log points in Indonesia and 22 log points in Kenya; analogous estimates show that urban productivity

3. Porzio (2017) argues that a model of worker sorting can explain a large share (roughly 40%) of intersectoral productivity gaps, considering agriculture as well as a range of non-agricultural sectors. Lagakos and Waugh (2013) similarly model how worker sorting across sectors could generate sectoral productivity differences in equilibrium.

4. The datasets—the Indonesian Family Life Survey and Kenya Life Panel Survey (henceforth "IFLS" and "KLPS")—are described in greater detail in what follows. In related work, Herrendorf and Schoellman (2018) employ cross-sectional microdata to assess sectoral differences in human capital.

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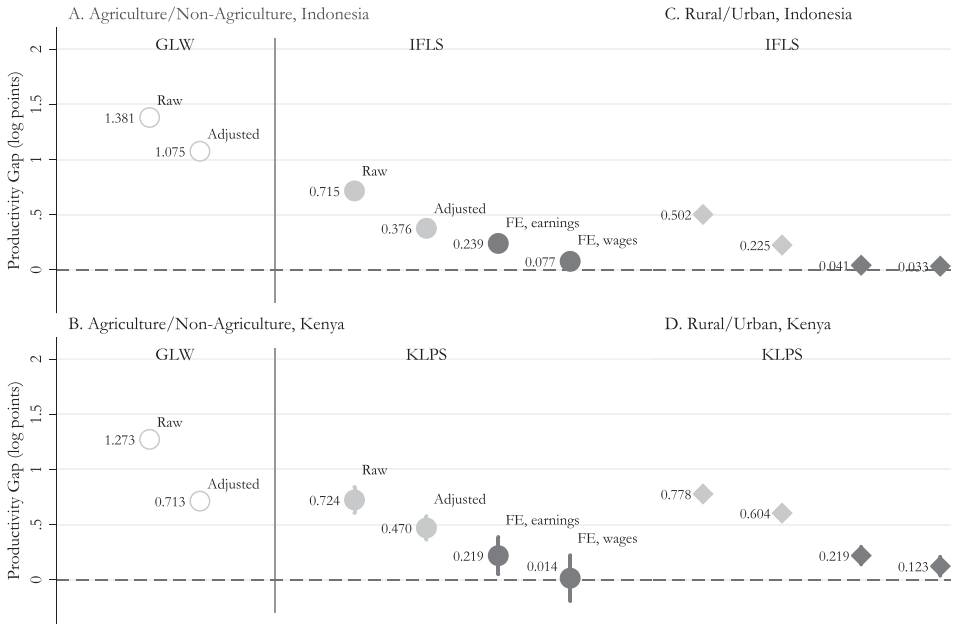


FIGURE 1. Productivity gap in total earnings. GLW refers to estimates from Gollin, Lagakos, and Waugh (2014), Online Appendix Table A.4. For comparability, the figure reports log transformed numbers from their columns (4) and (5) for Indonesia and Kenya, respectively. Symbols here represent point estimates, and vertical lines represent 95% confidence intervals. Panel A estimates from the IFLS come from Table 4, panel A: "Raw" is the mean difference estimate from column (1); "Adjusted" is the regression adjusted mean difference estimate from column (3); "FE, earnings" is the fixed-effects regression estimate of log earnings with hours controls in column (7); and "FE, wages" is the fixed-effects regression estimate of wages from column (8). Corresponding estimates from the KLPS come from Table 4, panel B. Estimates in panels C and D come from the same columns in Table 5, panels A and B, respectively. Note that the confidence intervals for the estimates from the IFLS are smaller than the size of the symbols and are therefore not visible.

gaps are also reduced substantially, to 4 log points in Indonesia and 22 log points in Kenya. The estimated gaps in GLW—as well as our estimated gaps obtained either without controlling for education and cognitive ability or including fixed effects—are an order of magnitude larger. Mirroring our results, Alvarez (2020) finds that wage gaps between agricultural and non-agricultural formal sector Brazilian workers are primarily due to differences in worker characteristics, rather than differential pay for similar workers across sectors. In contrast, Bryan, Chowdhury and Mobarak (2014) estimate positive gains in consumption (of roughly 30%) in the sending households of individuals randomly induced to migrate to cities within Bangladesh.[5] Using data from Tanzania and observing individuals at two points in time, Beegle, de Weerdt and Dercon (2011) estimate consumption gains of 36% among those who moved away

5. We return to a discussion of this study in the conclusion.

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from their origin area, yet estimating urban–rural gaps is challenging in their study setting, because only 138 individuals are observed in urban areas.

Bazzi et al. (2016) argue that cross-sectional estimates of productivity differences across rural areas within Indonesia are likely to overstate estimates derived from panel data using movers. They argue that human capital is location-specific and not necessarily readily transferable: Although they show that migrants eventually adapt to new settings, they argue that cross-sectional differences in productivity overstate causal impacts. Finally, although Bryan and Morten (2019) suggest there are substantial gains to removing migration frictions in Indonesia, gains are mainly associated with moves across region rather than urban residence per se.[6]

A limitation of the current study (at least in comparison to macro work) is its focus on two countries. This is due to the relative scarcity of long-run individual panel datasets in low-income countries that contain the rich measures necessary for our analysis. That said, the finding of similar patterns in two countries with large populations (250 million in Indonesia and 45 million in Kenya)—together with the similar results for formal sector wage gaps in Brazil (Alvarez 2020)—suggests some generalizability across different world regions.

Another important issue relates to the local nature of our estimates, namely, the fact that the fixed-effects estimates are derived from movers, those with productivity observations in both the non-agricultural and agricultural (or urban and rural) sectors. The inclusion of individual fixed effects addresses a large class of omitted variable biases that would cause one to incorrectly attribute the inherent productivity of certain individuals—their absolute advantage—to the sector itself. However, because sector itself is a choice, it is possible that productivity gains could be different among non-movers, an issue we discuss at length in Section 2 in what follows. A typical Roy (1951) model would suggest that the movers are also selected with respect to their individual-specific benefit—their comparative advantage—and thus our estimates identified from the population of movers would be larger than those for the population on average. Nonetheless, we acknowledge that our estimates would be downwardly biased if individuals who are especially well-suited to urban and non-agricultural work face particularly large barriers (e.g., credit constraints and information barriers) that are not compensated by the increase in expected earnings. Similarly, it is possible that very long-run and even intergenerational "exposure" to a sector could improve individual productivity due to skill acquisition. Indeed, we find that children born in urban areas score higher on cognitive tests than those born rural, even after controlling for parental education and cognitive performance. We return to issues of interpretation in the conclusion, including ways to reconcile our estimates with existing findings.

6. Related studies on selective migration include Chiquiar and Hanson (2005), Yang (2006), Kleemans (2016), and Rubalcava et al. (2008). Exploring international migration, Hendricks and Schoellman (2018) find that estimated returns are greatly reduced by the inclusion of individual fixed effects in panel data (by roughly 60%). Similarly, McKenzie, Gibson and Stillman (2010) show that cross-sectional estimates of the returns to international immigration (to New Zealand) exceed those using individual panel data or derived from a randomized lottery.

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The paper is organized as follows. Section 2 presents a conceptual framework for estimating sectoral productivity gaps, and relates it to the core econometric issue of disentangling causal impacts from worker selection and sorting. Section 3 describes the two datasets used; characterizes the distinctions between the non-agricultural and agricultural sectors, and urban versus rural areas; and presents evidence on individual selection between sectors. Section 4 contains the main empirical results on productivity gaps, and presents medium- and long-run dynamic effects. The final section presents alternative interpretations of the results, and concludes.

2. Theoretical Framework

We present a simple framework to disentangle explanations for the aggregate productivity gap across sectors. We consider both observable and unobservable components of human capital, and whether intrinsic preferences for sector may bias direct measurement of productivity gaps.

2.1. *Agricultural Productivity Gaps through the Lens of an Aggregate Production Function*

Following Hendricks and Schoellman (2018), we denote revenue productivity in sector s as $Q_s^r = K_s^{\alpha}(A_s H_s L_s)^{1-\alpha}$. Dropping subscripts for convenience, a representative firm in sector s solves

$$\max_{K,HL} K^{\alpha}(AHL)^{1-\alpha} - RK - ZHL$$

where R and Z represent returns per unit of physical capital K and a wage per efficiency unit of labor (comprised of the product of human capital per unit of labor, H, and labor quantity, L), respectively. Solving the first-order condition with respect to the labor aggregate yields

$$Z = (1-\alpha)\left(\frac{K}{Q^r}\right)^{\alpha/1-\alpha} A.$$

Though this wage equation does not represent a closed form inverse labor demand function per se, it captures how sector-specific wages depend on the inherent productivity of the sector and how intensely it uses capital. We follow Hendricks and Schoellman (2018) and treat the capital–output ratio as exogenously determined from the perspective of the worker. Finally, we assume that labor is perfectly substitutable, but workers supply different amounts of human capital.

An individual's income in sector s is given by $Y_{is} = Z_s H_i L_{is}$. Denoting logs in lower case, the average log-income gap across the non-agricultural (n) and agricultural (a) sectors is

$$\overline{y_n} - \overline{y_a} = \underbrace{(z_n - z_a)}_{\text{residual income gap}=\beta} + \underbrace{\left(\overline{l_n} - \overline{l_a}\right)}_{\text{labor supply gap}} + \underbrace{\left(\overline{h_n} - \overline{h_a}\right)}_{\text{human capital gap}}. \quad (1)$$

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The agricultural productivity gap is comprised of a labor supply gap, a human capital gap, and a productivity residual, β, the key parameter of interest. To the extent that there are no systematic compensating differentials, a nonzero β suggests that there are labor market frictions that prevent workers from switching sectors to drive marginal revenue products to be equalized.

We assume that individual human capital is not sector-specific and takes a Mincerian form, $H_i = \exp[\boldsymbol{x}_i'\boldsymbol{b} + \eta_i]$ where $\boldsymbol{x}_i$ is a vector of observed characteristics (e.g., years of schooling) with corresponding returns $\boldsymbol{b}$, and η_i represents unobserved skill. Substituting into the income equation yields a simple regression formulation of log income in sector s:

$$y_i = z_s + \beta D_i + l_i + \boldsymbol{x}_i'\boldsymbol{b} + \eta_i, \tag{2}$$

where D_i is an indicator for work in nonagriculture. This setup implies a unit coefficient on log hours, and estimation could be implemented in a specification with log wages ($w_i = y_i - l_i$) as the dependent variable without labor hours controls: $w_i = z_s + \beta D_i + \boldsymbol{x}_i'\boldsymbol{b} + \eta_i$. This arises because the sectoral gap is determined in aggregate: Holding all else constant, an individual's labor hours choices do not generate meaningful movements along the labor demand curve. However, in a richer model in which hours affect productivity (e.g., because of fatigue or limited available work activities), then hours worked could be flexibly included as a control in $\boldsymbol{x}_i$. We explore different regression specifications along these lines in Section 4; doing so also allows for more direct comparison with existing research, which features various productivity measures.

The agricultural productivity gap becomes

$$\overline{y_n} - \overline{y_a} = \beta + \left(\overline{l_n} - \overline{l_a}\right) + \left(\overline{\boldsymbol{x_n}} - \overline{\boldsymbol{x_a}}\right)'\boldsymbol{b} + \left(\overline{\eta_n} - \overline{\eta_a}\right). \tag{3}$$

It is evident that any differences in unobserved components of worker human capital will be absorbed into the residual wage gap here, and an ordinary least squares estimate of β will be biased.

There are two immediate approaches for obtaining more valid estimates of β. First, one can obtain a richer set of observable characteristics $\boldsymbol{x}_i$, reducing the scope for unobserved (to the econometrician) ability. Second, one can utilize panel data and estimate *within person* wage differences over time to purge the estimation of the time-invariant components of unobserved individual characteristics. Although our estimation explores both avenues, our preferred estimates use the second approach, using fixed-effects panel data estimation.

In a dynamic setting, the Mincerian human capital equation changes slightly to become $H_{it} = \exp[\boldsymbol{x}_i'\boldsymbol{b} + \eta_i + \omega_{it}]$. Here, η_i is again unobserved individual skill, and ω_{it} is a mean zero, individual, time-varying shock. An individual's time-invariant human capital (which we estimate in what follows as an individual fixed effect) is thus $\theta_i = \boldsymbol{x}_i'\boldsymbol{b} + \eta_i$. Equation (2) becomes

$$y_{it} = z_a + \beta D_{it} + l_{it} + \theta_i + \omega_{it} \tag{4}$$

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and the analog of equation (3) is

$$\overline{y_{nt}} - \overline{y_{at}} = \beta + \left(\overline{l_{nt}} - \overline{l_{at}}\right) + \left(\overline{\theta_n} - \overline{\theta_a}\right) + \left(\overline{\omega_{nt}} - \overline{\omega_{at}}\right). \quad (5)$$

Here, the time-varying components of human capital ω_{it}, are potential sources of omitted variable bias. Equation (4) is the key estimation equation; we explore potential limitations in what follows.

2.2. *Econometric Issues Related to Worker Selection and Heterogeneous Sectoral Effects*

In the previous section, for simplicity, we assumed that working in a sector had a constant effect on productivity for all individuals. In that framework, the dominant challenge in interpreting cross-sectional regression-adjusted comparisons as causal was absolute advantage: Omitted unobservable characteristics of human capital that lead to higher individual productivity regardless of sectoral choice. Our approach of utilizing individual fixed and focusing on the labor market experiences of sectoral switchers largely addresses those concerns.

However, it is well-known using the sample of switchers to infer population average effects can yield biased estimates in the presence of effect heterogeneity stemming from individual comparative advantages. In this section, we relax the constant effects assumption in a Roy (1951) model of self-selection, and explore issues related to inferring a population average sectoral effect when focusing on the subset of individuals who actually switch.

Here, we first specify human capital to be sector-specific and modify our Mincerian human capital equation to be $H_{ist} = \exp[\theta_{is} + \omega_{ist}]$. This formulation generates individual and sector-specific earnings—and correspondingly, incentives—to choose sectors. Modeling sectoral choice formally, worker utility in sector s is given by $v_{ist}(y_{ist})$, which is a function of the income that they would earn in that sector y_{ist}. Workers choose to work in nonagriculture if their utility for working in the sector is higher than in agriculture, $v_{int} - v_{iat} \geq 0$.

To explore self-selection bias, we further assume that net utility can be written nonparametrically as $v_{int}(y_{int}) - v_{iat}(y_{iat}) = (y_{int} - y_{iat}) + \xi_{it}$, where ξ_{it} is an idiosyncratic preference for working in nonagriculture meant to capture other benefits (e.g., a personal taste for urban non-agricultural work) and costs (e.g., pecuniary migration costs and nonpecuniary influences of past migration events) that are not already captured in productivity. We do not impose further restrictions on ξ_{it}: for instance, if there are substantial switching costs or experience effects, then ξ_{it} could be serially correlated; if there are compensating differentials, then ξ_{it} is not mean zero; and ξ_{it} may also correlate with other worker characteristics.

Substituting in the individual-specific productivity term from equation (4) (modified to allow for sector-specific human capital), the probability an individual chooses a particular sector (abstracting away from labor supply differences across

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sectors for parsimony) is given by

$$\Pr\left\{v_{int} - v_{iat} > 0\right\} = \Pr\{\beta + \left(\theta_{in} - \theta_{ia}\right) + \left(\omega_{int} - \omega_{iat}\right) + \xi_{it} > 0\}. \quad (6)$$

The possible bias here is classic simultaneity: We observe workers in the sector that benefits them the most. Unobserved wage innovations driven by h_{ist} simultaneously affect the worker's sectoral choice and her observed wage in that sector.

In a richer formulation of human capital with comparative advantage, the modified aggregate productivity gap (again abstracting away from labor supply differences) is

$$\begin{aligned}\overline{y_{nt}} - \overline{y_{at}} &= \beta + \boldsymbol{E}\left\{h_{int}|v_{int} > v_{iat}\right\} - \boldsymbol{E}\left\{h_{iat}|v_{iat} > v_{int}\right\} \\ &= \beta + \left(\boldsymbol{E}\left\{h_{iat}|v_{int} > v_{iat}\right\} - \boldsymbol{E}\left\{h_{iat}|v_{iat} > v_{int}\right\}\right) \\ &\quad + \boldsymbol{E}\left\{h_{int} - h_{iat}|v_{int} > v_{iat}\right\}. \end{aligned} \quad (7)$$

This may differ from our key parameter of interest, the true aggregate sectoral effect β, for two reasons. The first additional term on the right-hand side captures differences in agricultural productivity driven by absolute advantage, the focal point of the previous section, and largely addressed through the use of individual fixed effects. The second additional term captures bias due to possibly heterogeneous effects, which we turn to next.

Consider workers who are born in rural areas. Even if migrants and nonmigrants have the same productivity in agricultural work, if workers with a comparative advantage in nonagriculture are the ones who tend to migrate—that is $\mathbf{Cov}\{v_{int} - v_{iat}, y_{int} - y_{iat}\} > 0$—then a panel data fixed-effects analysis will continue to be biased upward because we observe urban wages only for those who benefit most.[7] This is analogous to generalizing an estimate of the "treatment effect on the treated" to an average treatment effect in the program evaluation literature, in a case where those who gain the most from a treatment are more likely to take up.

For the panel data fixed-effects analysis to instead generate a negatively biased estimate (understating the true sectoral effect β), a necessary condition is that $\mathbf{Cov}\{v_{int} - v_{iat}, y_{int} - y_{iat}\} < 0$. This is permitted by virtue of our unrestrictive formulation of nonwage utility ξ_{it}, and immediately implies that $\mathbf{Cov}\{\xi_{it}, y_{int} - y_{iat}\} \leq -1$. For this to hold, those with the most to gain from sectoral switches would not only need to face larger costs (i.e., $\mathbf{Cov}\{\xi_{it}, y_{int} - y_{iat}\} \leq 0$), but the costs associated with switching sector—whether pecuniary or psychological—would need to grow faster on average than the increase in benefits. In contrast, if nonwage preferences for work in a sector are positively correlated with, mean independent of, or only weakly negatively correlated with wage benefits, then panel data fixed effects estimates will again deliver upward biased estimates of the true sector premium.

It is an open empirical question how (nonwage) costs and preferences for sectoral work relate to sectoral wage gains across individuals, and we acknowledge that our data

7. Hendricks and Schoellman (2018) make the related point that their panel estimates of the returns to international migration are likely to be upper bounds on true effects.

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does not provide definitive evidence on this question. Simply put, we cannot rule out that there are some individuals, say in remote rural areas, who would both experience the largest wage gains from a move to an urban non-agricultural job (relative to rural agriculture), but also face especially high personal barriers to movement. Of course, to the extent that some of these costs simply capture individual disutility from working in an urban or non-agricultural environment, then this does beg the question of whether extreme policies would need to be enacted in order to achieve labor reallocation across sectors, and whether such policies would even be welfare improving.

A further consideration with the fixed-effects estimates is the fact that they are identified on the subset of workers for whom we observe earnings in both sectors. To be clear, our main estimates do not rely on data from individuals who do not move at all nor those who migrated prior to entering the labor market. Yet owing to the long-term nature of our datasets, we are able to observe many individuals in multiple sectors over the course of their entire work histories.

Finally, much of the previous discussion has implicitly focused on workers born in rural areas, which is the case in our Kenya data where the entire population lived in rural areas at baseline. However, the Indonesian data also features individuals born in urban areas and sorting in both directions. By a parallel logic to mentioned previously, it is conceptually possible to observe a non-agricultural (or urban) premium every time an individual selects into nonagriculture (urban areas) and an agricultural (rural) premium every time an individual selects into agriculture (rural areas). The resulting estimates would then serve as bounds on the true average effect.

The Indonesian dataset we use provides a testbed to understand the role of these biases, especially in terms of estimating the urban–rural gap. In the spirit of Young's (2013) observation that migrants flow in both directions, the data allow us to condition on individual birth location and measure the dynamic impacts on wages after migration. To the extent that it holds, the bounding argument mentioned previously predicts that the estimated urban–rural productivity gap would be larger when estimated for movers from rural to urban areas than it is when estimated for movers from urban to rural areas. We take this prediction to the data and find patterns consistent with it in what follows. This selection model implies that the true sectoral productivity gap in Indonesia is bounded by these two estimates, generated by movers in each direction.

3. Data

This paper uses data from Indonesia and Kenya. At 250 million, the Southeast Asian country of Indonesia is the world's fourth most populous, and Kenya is among the most populous countries in Sub-Saharan Africa, with 45 million people. They are fairly typical of other low-income countries with respect to the labor share in agriculture, agricultural productivity gaps using national accounts data, and relationships between these variables and national income levels.[8]

8. See Online Appendix Figures A.1 and A.2 based on data from GLW. In both figures, the values for both Kenya (KEN) and Indonesia (IDN) are close to the best fitting regression line.

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The high tracking rates of the datasets we employ allow us to construct multiyear panels of individuals' location decisions. Moreover, both datasets include information on formal and informal sector employment. The latter is difficult to capture in standard administrative data sources, yet often accounts for a large share of the labor force in low-income countries.

3.1. Indonesia

We employ data from five rounds of the IFLS, collected between 1993 and 2015 (Strauss, Witoelar and Sikoki 2016). The survey is representative of 83% of the country's population who lived in 13 of the 27 provinces that existed in 1993. Although the original 1993 sample consisted of 22,347 individuals, the inclusion of additional members from split-off households during subsequent rounds results in a sample of 58,337 individuals. Attrition is often high in panel data; however, with an intensive focus on respondent tracking, the IFLS is unusually well-suited to study migration. Re-contact rates between any two rounds are above 90%, and 87% of original households were contacted in all five rounds (Strauss et al. 2016).[9]

Detailed current and previous employment data (for primary and secondary employment) were collected during each IFLS survey round, allowing us to create up to a 28-year annual individual employment panel (from 1988 to 2015).[10] Respondents are asked to include any type of employment, including paid labor and self-employment, temporary work, work on a family-owned farm or nonfarm business, and unpaid family work. In addition to labor earnings and profits, individuals are asked to estimate the value of other benefits; our main earnings measure is the sum of all labor earnings, profits, and benefits.[11]

Individuals are asked to describe the sector of employment for each job. The single largest sector is "agriculture, forestry, fishing, and hunting": 31% of individuals report it as their primary employment sector, and 50% have secondary jobs in this sector. Agricultural employment is primarily rural: 47% versus 11% of rural and urban individuals, respectively, report working primarily in agriculture (Table 1, panel A). Other common sectors are wholesale, retail, restaurants, and hotels (22% of main employment); social services (22%); manufacturing (14%); and construction (5%). These non-agricultural sectors are all more common in urban areas. Men are more likely than women to work in agriculture (35% versus 23%) and less likely to work in wholesale, retail, restaurants, hotels, and social services. Smaller male-dominated sectors include construction (7% of male employment versus 0.7% for females) and transportation, storage, and communications (6% versus 0.4%).

In the analysis, we employ an indicator variable for non-agricultural employment, which equals 1 if a respondent's main employment is not in agriculture. The main

9. Thomas et al. (2012) contains a detailed discussion of tracking and attrition in the IFLS.

10. Employment status and sector are available for each year, but earnings were collected only for the current job in the fourth and fifth data collection rounds.

11. De Mel, McKenzie, and Woodruff (2009) argue that self-reported profits give a more accurate depiction of firm profits in microenterprises than reconstructed measures.

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TABLE 1. Nonagriculture/agriculture and urban/rural.

(A) Indonesia (main analysis sample)			
	Rural	Urban	Total
Agriculture	46.9%	10.7%	29.4%
Nonagriculture	53.1%	89.3%	70.6%
Number of observations	133,726	125,019	258,745
(B) Kenya (main analysis sample)			
	Rural	Urban	Total
Agriculture	26.0%	5.4%	15.2%
Nonagriculture	74.0%	94.6%	84.8%
Number of observations	61,750	68,572	130,322
(C) Kenya (12 months with subsistence agricultural module)			
	Rural	Urban	Total
Agriculture	59.1%	9.1%	40.6%
Nonagriculture	40.9%	90.9%	59.4%
Number of observations	27,301	16,029	43,330

Notes: Panel A reports summary statistics from the IFLS, and panels B and C present data from the KLPS; both are described in more detail in Section 3. Panel A shows the main Indonesian analysis sample of 258,745 individual-year observations, for individuals aged 15 and above for whom earnings measures are available. As described in Section 3, all reported wage employment, self employment, and subsistence agricultural employment is included. Panel B shows the main Kenyan analysis sample of 134,221 individual-month observations of individuals aged 16 and above for whom earnings measures are available. As described in Section 3, wage and self-employment are included for all years, including agricultural wage labor and agricultural self-employment if annual revenues exceeded 40,000 Ksh, and subsistence agricultural data is included from the 12 months preceding the KLPS 3 interview date if the respondent is both the main decision maker and the only supplier of labor in agriculture. Panel C shows data from the 12 months where subsistence agriculture data are available and counts all agricultural activities: including when the person is not the main decision maker and when others work on the agricultural activity; in the case of the latter, the agricultural productivity is weighted by the share of hours that the individual supplies. Each cell reports the percentage of observations by agricultural and non-agricultural sector, and by rural and urban area. In both the IFLS and KLPS, individuals are characterized by the sector of their main employment. The urban indicator from the IFLS is obtained from survey responses to the question: "Is the area you live in a village, a town or a city?" If the person reports living in a town or city, the urban indicator variable equals 1. For the KLPS, the urban indicator equals 1 if the person reports living in a large town or city. Please see the text in Section 3.2 for further details on this classification. The list of Kenyan urban areas and frequency of occurrence in the panel are given in Online Appendix Table A.5.

analysis sample includes all individuals who are employed and have positive earnings and positive hours worked to ensure that the principal outcome variables of interest, log earnings and log wages, are defined. The sample includes 31,537 individuals and 258,745 individual-year observations.[12] In addition to studying earnings, we explore

12. The panel is unbalanced due to attrition, death, and to limiting observations to respondents at least 15 years old.

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consumption (in Online Appendix C) to get a broader sense of overall welfare and total income.

Data were collected on the respondent's location at the time of the survey, and all rounds of the IFLS also collected a full history of migration within Indonesia, including all residential moves across subdistricts ("kecamatan") that lasted at least 6 months (so, for example, seasonal migration is excluded). We combine data across IFLS rounds to construct a 28-year panel, from 1988 to 2015 with annual information on the person's location, in line with the employment panel.[13] Figure 2, panel A presents a map of Indonesia with each dot representing an IFLS respondent's residential location. Although many respondents live on Java, we observe considerable geographic coverage throughout the country.

We utilize a survey-based measure of urban residence: If the respondent reports living in a "village," we define the area to be rural, whereas they are considered urban if they answer "town" or "city." We present the correspondence between urban residence and employment in the non-agricultural sector in Table 1, panel A. In 71% of individual-year observations, people are employed in the non-agricultural sector, and in 48% of the observations, they live in urban areas. One can see that a substantial portion of rural employment is in both agriculture and non-agricultural work, whereas urban employment is almost exclusively non-agricultural, as expected.

Given the migration focus of the analysis, it is useful to report descriptive statistics both for the main analysis sample, as well as separately for individuals in four mutually exclusive categories (Table 2, panel A): Those who always reside in rural areas during IFLS ("Always Rural"), those who were born in a rural area but move to an urban area at some point ("Rural-to-Urban Migrants"), those who are "Always Urban," and finally, the "Urban-to-Rural Migrants" (born urban but move to a rural area at some point). As discussed previously, the fixed-effects analysis is driven by individuals who move between sectors during the sample period.

In the main IFLS analysis sample, 88% of adults had completed at least primary education, and more than a third had completed secondary education, whereas tertiary education remains quite limited, at 11%. Among those who are born in rural areas in columns (2) and (3) (of Table 2, panel A), we see that migrants to urban areas are highly positively selected in terms of both educational attainment and cognitive ability, with Raven's Progressive Matrices exam scores more than 0.2 standard deviation units higher among those who migrate to urban areas, a meaningful effect.[14] There is also evidence of migration rates differing across gender.

These relationships are presented in a regression framework in Table 3, column (3), and the analogous relationships for moves into non-agricultural employment are

13. See Kleemans (2016) and Kleemans and Magruder (2018) for information on IFLS employment and migration data.

14. Raven's Matrices were administered to a subset of individuals in IFLS 3, 4, and 5, namely, those aged 7–24 years old. Raven's Matrices test is designed to capture fluid intelligence.

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(A) Indonesia Family Life Survey



(B) Kenya Life Panel Survey

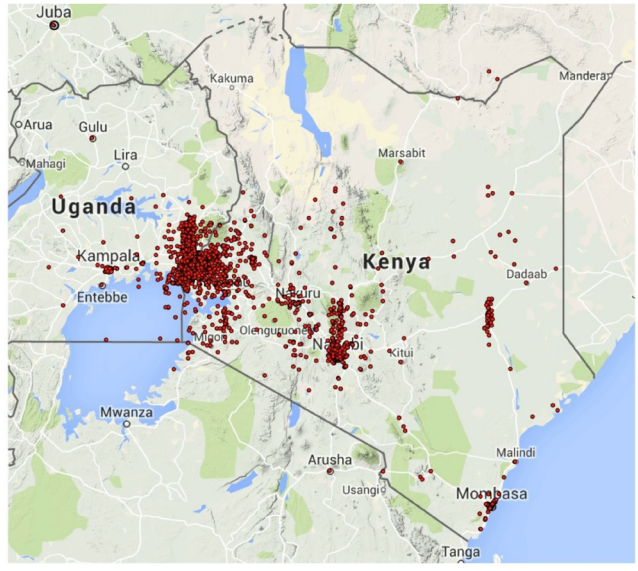


FIGURE 2. Sample areas. Panel A shows the residential locations of individuals during the 1988–2008 sample period of rounds 1–4 of the IFLS. For the Kenyan sample, panel B shows individuals' residential locations during the 1998–2014 sample period that was collected during rounds 2 and 3 of the KLPS. Individuals living outside of Kenya are dropped from the analysis. The location information of both datasets are described in more detail in Section 3.

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TABLE 2. Summary statistics.

	All	Always rural	Rural-to-urban migrants	Always urban	Urban-to-rural migrants	Observations
(A) Indonesia						
	$N = 31{,}537$	$N = 11{,}927$	$N = 9{,}881$	$N = 7{,}226$	$N = 2{,}492$	
Primary education	0.875	0.793	0.900	0.966	0.907	31,537
	[0.330]	[0.405]	[0.300]	[0.182]	[0.291]	
Secondary education	0.397	0.246	0.391	0.629	0.469	31,537
	[0.489]	[0.431]	[0.488]	[0.483]	[0.499]	
College	0.109	0.051	0.101	0.202	0.148	31,537
	[0.312]	[0.220]	[0.301]	[0.402]	[0.356]	
Female	0.432	0.419	0.433	0.461	0.400	31,537
	[0.495]	[0.493]	[0.496]	[0.498]	[0.490]	
Raven's Z-score	0.000	−0.183	0.048	0.194	0.107	23,214
	[0.923]	[0.926]	[0.917]	[0.870]	[0.926]	
(B) Kenya						
	$N = 4{,}718$	$N = 1{,}603$	$N = 3{,}115$			
Primary education	0.734	0.638	0.783			4,718
	[0.442]	[0.481]	[0.412]			
Secondary education	0.352	0.240	0.411			4,718
	[0.478]	[0.427]	[0.492]			
College	0.035	0.012	0.046			4,718
	[0.183]	[0.108]	[0.210]			
Female	0.521	0.519	0.522			4,718
	[0.500]	[0.500]	[0.500]			
Raven's Z-score	0.051	−0.142	0.149			4,452
	[0.985]	[0.978]	[0.974]			

Notes: Panel A reports summary statistics from the IFLS and panel B reports summary statistics from the KLPS. Sample standard deviations reported in brackets below sample means. The sample is limited to respondents who report age, gender, and years of education and have at least one person-time observation that has income, hours, location of residence, and sector of occupation. In panel A (Indonesia), "Rural-to-urban migrants" are individuals born in rural areas and are observed in urban areas in our sample with data on wages, hours, and sector. "Urban-to-rural migrants" are defined similarly. In panel B (Kenya), all individuals are born rural; migrants are those who have subsequent observations with information on income, hours, and sector in urban areas. Rows correspond to the fraction within each column who have completed primary education, secondary education, and college; the fraction female; and the average score from a Raven's Matrices exam, normalized to be mean zero and standard deviation 1.

also evident (column (1)).[15] Importantly, the relationship between higher cognitive ability and likelihood of migrating to urban areas holds even conditional on schooling attainment and demographic characteristics, at 99% confidence. This indicates that

15. Online Appendix Tables A.1 and A.2 present a full set of regression results, where each control is first considered in isolation, and the full set of controls are considered together in column (6).

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TABLE 3. Correlates of employment in nonagriculture and urban migration.

	Dependent variable: non-agricultural employment		Dependent variable: urban migration	
	(1) Indonesia	(2) Kenya	(3) Indonesia	(4) Kenya
Primary education	0.212***	0.103***	0.124***	0.086***
	(0.013)	(0.014)	(0.013)	(0.019)
Secondary education	0.131***	0.045***	0.090***	0.099***
	(0.007)	(0.008)	(0.009)	(0.017)
College	0.051***	0.015*	0.039**	0.114***
	(0.007)	(0.008)	(0.016)	(0.028)
Female	0.082***	0.031***	0.036***	0.018
	(0.006)	(0.009)	(0.008)	(0.014)
Raven's Z-score	0.036***	0.021***	0.047***	0.031***
	(0.004)	(0.005)	(0.004)	(0.008)
Constant	0.514***	0.796***	0.305***	0.548***
	(0.013)	(0.013)	(0.012)	(0.017)
Observations	16,041	4,452	16,041	4,452

Notes: See Table 2 for sample restrictions and row variable definitions. The dependent variable in the first two columns is an indicator for being ever being employed in nonagriculture and in the last two columns the dependent variable is an indicator for being an urban migrant. All regressions are estimated on individuals who are born rural. Robust standard errors are in parentheses.
$*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

sorting on difficult-to-observe characteristics is relevant in understanding sectoral productivity differences.

It is worth noting that if we ignore migrants, individuals who are born and remain in urban areas are far more skilled than those who stay in rural areas. "Always Urban" individuals in Table 2, panel A score nearly 0.4 standard deviation units higher on Raven's Matrices and have more than double the rate of secondary schooling and four times the rate of tertiary education relative to "Always rural" individuals. The urban-to-rural migrants in Indonesia are also negatively selected relative to those who remain urban, consistent with Young (2013).[16]

3.2. Kenya

For the analysis in Kenya, we employ the KLPS, which tracks over a 10–15 year period 8,999 individuals who attended primary school in western Kenya in the late 1990s and early 2000s. These individuals are a representative subset of students enrolled in over 200 rural primary schools in Busia district, schools that participated in one of two school-based randomized interventions: A scholarship program for upper primary

16. Online Appendix Tables A.3 and A.4 report analogous results among those individuals born in urban areas.

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school girls that took place in 2001 and 2002 (Kremer, Miguel, and Thornton 2009) and a deworming treatment program for primary school students during 1998–2002 (Miguel and Kremer 2004). According to the 1998 Kenya Demographic and Health Survey, 85% of children in Western Province aged 6–15 were enrolled in school at that time, and Lee et al. (2015) show that this area is quite representative of rural Kenya as a whole in terms of socioeconomic characteristics. To date, three rounds of the KLPS have been collected (2003–2005, 2007–2009, 2011–2014).

KLPS data collection was designed with attention to minimizing bias related to survey attrition. Sample individuals who had left the original study area were tracked throughout Kenya (as well as into neighboring Uganda and beyond, although we exclude international migrants from the present analysis).[17] The effective tracking rate for each KLPS round is roughly 85%.[18]

Similar to the IFLS, the KLPS includes information on educational attainment, labor market participation, and migration choices. Employment data was collected in paid employment and self-employment modules, designed to capture both formal and informal employment. Most individuals were quite young (typically teenagers) during data collection for KLPS Round 1, and few had paid employment or self-employment to report. Full employment histories, including more detailed questions, were collected during Rounds 2 and 3, and it is from these rounds that we draw the data on individual earnings, hours worked, and wages used in the present analysis. Detailed information on agriculture productivity was collected only in the third round, and we present analysis including this measure for those interviewed in KLPS Round 3.[19]

17. The results presented in what follows are robust to the inclusion of international migrants (not shown).

18. Respondents were sought in two separate "phases" of data collection: The "regular tracking phase" proceeded until over 60% of respondents had been surveyed, at which point a representative subset of approximately 25% of the remaining sample was chosen for the "intensive tracking phase" (and remaining unfound individuals no longer sought). These "intensive" individuals receive roughly four times as much weight in the analysis, to maintain representativeness with the original sample. Baird, Hamory and Miguel (2008) describe this methodology in detail.

19. The Kenya agricultural productivity data deserve some discussion. Whenever total household annual agricultural sales were at least moderate, exceeding 40,000 Kenyan shillings (approximately US$400–500), full agricultural production and profit information was collected in the self-employment module and included in the present analysis. Agricultural paid employment is also common, and these data are always included. Some questions on subsistence agricultural production were collected in KLPS Rounds 1 and 2, but these are insufficient to create an individual productivity measure, but more detailed information on agricultural productivity (measured for the 12 months preceding the interview) is contained in Round 3, and this is included in the present analysis. To create a measure of individual productivity comparable with other sectors, we focus on agricultural activities (e.g., growing a particular crop) in which the respondent provided all reported labor hours; we also restrict attention to activities in which the respondent reports being the main decision-maker, because it seems likely that they are most knowledgeable about such activities (although results are not sensitive to this restriction). The profit in an agricultural activity is the sum of all crop-specific production—valued either through actual sales or at the relevant crop price if consumed directly—minus all input costs and hired labor costs. The individual wage divides this net profit by the labor hours the respondent supplied to the activity. We show in what follows that estimates are robust to alternative individual agricultural productivity measures, including the exclusion of subsistence agriculture data.

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KLPS respondents reported industry for all paid employment and self-employment. Most individuals are engaged in relatively low-skilled work: the most common industry for paid employment is services, at 57% overall and 74% for females (with many women in domestic services). In rural areas, the most common industries for paid employment are services and agriculture (50% and 21%, respectively), whereas in urban areas they are services, and manufacturing and construction (62% and 11%). The largest self-employment industries are retail and services (41% and 25%).[20]

KLPS respondents provide a history of residential locations because their last interview, and this data includes residential district, town, and village, allowing us to classify individuals who self-report living in towns and cities as urban residents (following the IFLS classification methodology). The KLPS includes information on all residential moves that lasted at least 4 months in duration, a slightly more permissive definition than in the IFLS, and we are able to construct a monthly residential panel from March 1998 to October 2014.[21] Combined with the retrospective labor productivity data, the main analysis sample is a monthly panel with 130,322 individual-month observations for 4,718 individuals.

Figure 2, panel B presents a map of Kenya, with each dot representing a respondent residential location during 1998–2014. Most residences in western Kenya are in Busia district (where the sample respondents originally resided), with substantial migration to neighboring areas as well as to cities. Online Appendix Table A.5 presents the list of main towns and cities, and shows that more than 70% of urban residential moves are to Kenya's five largest cities, namely, Nairobi, Mombasa, Kisumu, Nakuru, and Eldoret. Men are slightly more likely than women to report migrating for employment reasons (60% of moves compared to 55% for females) whereas women are more likely to migrate for family reasons, including marriage (13% versus 1% for men). A smaller share of moves (approximately 6%) are for education.

Summary statistics on employment sector and urban residence for KLPS respondents are presented in Table 1. Panel B presents data for the main analysis sample; as described previously, this contains subsistence agricultural information where available (from KLPS-3). The employment share in agriculture is much higher in rural areas (26.0%) than urban (5.4%), as expected, but the share in rural areas is lower than expected, likely because subsistence agricultural activities were not captured in earlier KLPS rounds. For a more complete portrait, panel C focuses on data from the 12 months prior to the KLPS-3 survey, which contains detailed information on

20. For paid employment, respondents also report occupation, and these tell a similar story. The most common occupations fall in the "unskilled trades" category (32%), followed by "skilled & semi-skilled trades" (19%), "retail and commercial" (18%), "professionals" (16%), and "agriculture" (15%). Agricultural paid employment is more common for men (20%) than women (6%), and as expected, agricultural employment is far higher in rural than urban areas (29% versus 5%). Common urban occupations are "unskilled trades" (37%), "skilled & semi-skilled trades" (22%), and "retail and commercial" (20%).

21. Similar to the IFLS, the panel is unbalanced due to attrition, death, and inclusion of individuals 16 and older.

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subsistence agriculture, and here the agricultural employment share in rural areas is much higher.

Recall that the Kenya sample is all rural at baseline (respondents were originally attending rural schools). Similar patterns emerge regarding positive selection into urban migration, with educational attainment and normalized Raven's Matrices scores both far higher among those who migrate to cities (Table 2, panel B). In particular, there is a raw gap of 0.3 standard deviation units in Raven's Matrices scores between urban migrants and those who remain rural. Table 3 reports these patterns in terms of regression estimates, for urban migration (column (4)) and employment in non-agricultural work (column (2)), respectively. As with Indonesia, controlling for educational attainment and gender, the Raven's score remains strongly positively correlated with urban migration (at 99% confidence).[22]

4. Results

4.1. Main Agricultural and Urban Productivity Gap Estimates

GLW estimate raw and adjusted agricultural productivity gaps of 138 and 108 log points in Indonesia, respectively (Figure 1, panel A). The estimate of this raw gap from the IFLS is somewhat smaller at 72 log points (Table 4, panel A). The most straightforward explanation for this discrepancy is an issue of measurement. GLW observe that, in an analysis of 10 countries, the average agricultural productivity gap was 17 log points smaller when estimated in Living Standards Measurement Study data that is similar to the IFLS, and which is more likely to capture earnings in informal employment.[23] That said, the raw gap we estimate in the IFLS remains substantial.

Inclusion of control variables similar to those used by GLW to adjust macro data gaps—in particular, the inclusion of log hours, log hours squared, and time-fixed effects (in column (2)), and then additionally an indicator for female, years of education, and years of education squared (in column (3))—reduces the estimated agricultural productivity gap in the IFLS to 60 and 38 log points, respectively (Table 4, columns (2) and (3)). Estimating on the subsample for which we have scores from Raven's Matrices tests, the gap is reduced slightly, although note the smaller sample size in this case (column (4)).

Limiting the analysis to those who have productivity measurements at some point in both agricultural and non-agricultural employment further reduces the productivity gap (column (5)), suggesting that selection on unobservable characteristics may be meaningful.

22. Online Appendix Tables A.1 and A.2 present a full set of regression results, where each control is first considered in isolation, and the full set of controls are considered together in column (6).

23. This comes from log transformed values from the "Average" row of GLW, Table 4, i.e.: $\ln 2.6 - \ln 2.2 = 0.167$.

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TABLE 4. Non-Agricultural/agricultural gap in earnings.

(A) Indonesia

	Dependent variable: log earnings								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8) Log wage	(9) Log real wage
Non-Agricultural employment	0.715***	0.604***	0.376***	0.364***	0.233***	0.299***	0.239***	0.077***	0.075***
	(0.013)	(0.012)	(0.012)	(0.014)	(0.018)	(0.018)	(0.017)	(0.020)	(0.020)
Log hours		0.564***	0.434***	0.454***	0.290***		0.350***		
		(0.016)	(0.015)	(0.017)	(0.038)		(0.019)		
Log hours squared		−0.023***	−0.009***	−0.013***	0.019***		−0.007**		
		(0.003)	(0.003)	(0.003)	(0.007)		(0.003)		
Female			−0.440***	−0.439***	−0.475***				
			(0.010)	(0.012)	(0.028)				
Years of education			0.015***	0.002	0.027***				
			(0.004)	(0.005)	(0.009)				
Years of education squared			0.004***	0.004***	0.003***				
			(0.000)	(0.000)	(0.001)				
Normalized Raven's				0.065***					
				(0.006)					
Normalized Raven's squared				0.015***					
				(0.005)					
Individual fixed effects	N	N	N	N	N	Y	Y	Y	Y
Time fixed effects	N	Y	Y	Y	Y	Y	Y	Y	Y
Switchers only					Y				
Number of observations	258,745	258,745	258,745	196,354	48,479	258,745	258,745	258,745	258,580
Number of individuals	31,537	31,537	31,537	23,214	3,907	31,537	31,537	31,537	31,530

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TABLE 4. Continued

(B) Kenya

	Dependent variable: log earnings								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8) Log Wage	(9) Log real wage
Non-Agricultural employment	0.724***	0.471***	0.470***	0.479***	0.272***	0.333***	0.219**	0.014	0.003
	(0.060)	(0.056)	(0.054)	(0.055)	(0.094)	(0.086)	(0.086)	(0.106)	(0.106)
Log hours		0.350**	0.254*	0.268*	0.283		0.218		
		(0.170)	(0.152)	(0.153)	(0.323)		(0.242)		
Log hours squared		0.014	0.017	0.016	0.008		0.015		
		(0.019)	(0.017)	(0.017)	(0.039)		(0.026)		
Female			−0.491***	−0.467***	−0.535***				
			(0.034)	(0.036)	(0.110)				
Years of education			0.007	−0.002	−0.012				
			(0.035)	(0.036)	(0.117)				
Years of education squared			0.004**	0.004**	0.004				
			(0.002)	(0.002)	(0.007)				
Normalized Raven's				0.072***	0.011				
				(0.021)	(0.069)				
Normalized Raven's squared				−0.045**	−0.138**				
				(0.018)	(0.070)				
Individual fixed effects	N	N	N	N	N	Y	Y	Y	Y
Time fixed effects	N	Y	Y	Y	Y	Y	Y	Y	Y
Switchers only					Y				
Number of observations	130,322	130,322	130,322	124,481	14,345	130,322	130,322	130,322	130,251
Number of individuals	4,718	4,718	4,718	4,452	324	4,718	4,718	4,718	4,715

Notes: Panel A uses data from rounds 1–5 of the IFLS, described in Section 3. Panel B uses data from rounds 2–3 of the KLPS, also described in Section 3. The dependent variable in columns (1)–(6) is log earnings, which are the combined earnings from wage and self-employment, reported in Indonesian Rupiah for Indonesia, and Kenyan Shillings for Kenya. If an individual has multiple jobs in the same time period, earnings from all employment are included. The dependent variable in column (7) is log wage, which is obtained by dividing log earnings by total hours worked. The dependent variable in column (8) is log wage adjusted for differences in prices between urban and rural areas, see Section 4 for details on this deflator. The covariate "Non-Agricultural employment" is an indicator variable which equals 1 if the person main employment is in the non-agricultural sector. The covariate log hours sums up hours worked in all employment. The sample size in column (4) is smaller in panel A because the Raven's test was administered only for a subset of the sample. The sample size in column (5) is smaller because it only includes "switchers" who have at least one observation in both the non-agricultural and agricultural sector. Each regression in columns (2)–(8) include quadratic controls for age. All regressions are clustered at the individual level. Robust standard errors are in parentheses.
$*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

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In column (6), we include individual fixed effects without hours controls, which leads to a productivity gap of 30 log points. In the last three columns, we account for hours in two different ways. Equation (4) in the conceptual framework implies a coefficient estimate of 1 on the hours measure, suggesting a regression specification with log wages as the dependent variable, but as noted previously, controlling for hours worked could be appropriate under some conditions. When we control for a quadratic function in hours worked (column (7)), the gap falls to 24 log points and the data reject the assumption of a coefficient estimate of 1 on hours worked, instead suggesting a nonlinear relationship. By itself, this nonlinearity suggests that individuals' productivity may change depending on how many hours they work and thus could usefully be incorporated into the vector of controls. However, there are other explanations for the departure from 1 on the hours worked coefficient that make this specification less attractive; for instance, a coefficient less than 1 may also reflect attenuation bias due to measurement error in the hours worked variable, or bias driven by endogenous individual hours worked decisions.

As a result, we also present the log wages specification implied by the conceptual framework. Using the wage (defined as total earnings divided by hours worked) as the dependent variable reduces the gap: the coefficient estimate falls to 0.077 (s.e. 0.020) in column (8), and further to 0.075 when considering the real log wage (adjusting for urban prices, column (9)).[24]

We follow a similar approach for Kenya, where the raw agricultural productivity gap falls from 72 log points to 47 with the inclusion of controls (Table 4, panel B, columns (1)–(4)), and to 22 log points when including an individual fixed effect in column (7). Using the hourly wage measure reduces the gap to 1.4 log points (column (8)), it falls further when adjusted with an urban price deflator (column (9)), and neither wage estimate is significant at traditional levels.[25]

Comparing column (1) (the raw gap) to column (7) in Table 4, the agricultural productivity gap is reduced by 67% in Indonesia and by 70% in Kenya. The upper end of the 95% confidence intervals are consistent with meaningful productivity gains to non-agricultural employment. That said, even these values remain far lower than the 108 and 71 log point effects that GLW estimate for Indonesia and Kenya, respectively, once they condition on observable labor characteristics (namely, hours worked and educational

24. The IFLS provides spatial deflators separately for urban and rural areas in each province, which are based on a representative basket of goods consisting of food and nonfood expenditure. On average, they show that prices in urban areas are 10% higher than in rural areas.

25. The KLPS includes contemporaneous price surveys in urban and rural markets located near survey respondent residences (in particular, urban market price data in Nairobi, Mombasa, and Busia Town, and rural markets in western Kenya), using a standard basket of food and nonfood goods across both sets of locations. This is used to calculate an urban price deflator, which is equal to the relative cost of purchasing this basket of goods in an urban area compared to a rural area. Specifically, the urban price deflator was calculated by measuring each item's total share of consumption in rural areas, and then multiplying by the average urban price for that item divided by the average rural price. These shares are then summed across all items, creating an urban price deflator. On average, the deflator suggests that prices in urban areas are 8% higher than in rural areas.

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attainment). These results for Indonesia and Kenya are presented graphically in Figure 1, panels A and B and compared to GLW's estimated productivity gaps.[26]

Table 5 presents the closely related exercise of estimating the labor productivity gap between residents of urban and rural areas. Although the existing empirical literature has sometimes conflated these two gaps, Table 1 shows that employment in rural areas is not exclusively characterized by agriculture. To the extent that residential migration is costlier than shifting jobs (but not homes), and the urban and non-agricultural earnings premia are related but distinct parameters, an urban earnings premium might be even more pronounced than the non-agricultural wage earnings premium.

The raw gap reported in column (1) of Table 5 (panel A) is 50 log points for Indonesia. Similar to the agricultural productivity gap, the urban–rural productivity gap falls when additional explanatory variables are added in columns (2)–(4), but remains substantial and statistically significant. Focusing the analysis only on those who have earnings measures in both urban and rural areas (column (5)) leads to a further reduction. Finally, the urban-rural earnings gap falls to 4.1 log points with the inclusion of individual fixed effects in Indonesia (column (7)) and further to 3.3 log points for the log wage measure (column (8)). Raw urban productivity differences are much larger for Kenya (78 log points) than Indonesia. Adjusting for individual fixed effects, urban productivity estimates for Kenya are still larger at 22 log points for earnings and 12.3 log points for wages. Thus, the productivity gap in earnings falls by 92% in Indonesia and by 72% in Kenya with the inclusion of individual fixed effects, whereas the reduction of the productivity gap measured in wages is 97% and 83% for Indonesia and Kenya, respectively. Once again, these results are summarized in Figure 1 (panels C and D). Urban productivity gaps in real wage terms (that account for higher urban prices) are further reduced in both countries (column (9)).[27] Online Appendix Table A.7 shows similar findings when we focus on the agricultural productivity gap in rural areas only.

The selection model presented in Section 2 predicts that estimated productivity gaps would be higher among rural-to-urban migrants than for urban-to-rural migrants, given plausible individual sorting patterns. Table 6 explores this hypothesis in Indonesia by separately conditioning on birth location; panel A contains those born rural and panel B those born in urban areas. The same pattern of declining urban productivity gaps in each subsample is observed as additional controls are included (columns (1)–(3)). In the log earnings specification in column (2), productivity gaps are indeed larger for those born in rural areas, as predicted by sorting (p-value < 0.01), whereas in the

26. Similar patterns hold when using alternative definitions of non-agricultural employment, namely, classifying simultaneous work in both sectors as agriculture or nonagriculture (Online Appendix Table A.6), which we return to in what follows.

27. In order to explore whether those working in agriculture and rural areas face intensive-margin constraints in labor demand, we show the difference in hours worked across sector and region in Online Appendix Tables A.8 and A.9. Those working in nonagriculture and those living in urban areas work slightly more hours in both countries, and the gap is only slightly diminished when adding controls and fixed effects in Online Appendix Table A.8, although the gap is decreased substantially in Online Appendix Table A.9. In line with our model and the literature, we divide earnings by hours worked to obtain wages.

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TABLE 5. Urban/rural gap in earnings.

(A) Indonesia

	Dependent variable: log earnings								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8) Log wage	(9) Log real wage
Urban	0.502***	0.422***	0.225***	0.200***	0.090***	0.043***	0.041***	0.033**	−0.060***
	(0.011)	(0.011)	(0.009)	(0.011)	(0.015)	(0.013)	(0.012)	(0.014)	(0.014)
Log hours		0.536***	0.414***	0.433***	0.373***		0.343***		
		(0.016)	(0.015)	(0.016)	(0.031)		(0.019)		
Log hours squared		−0.012***	−0.002	−0.005*	0.007		−0.005		
		(0.003)	(0.003)	(0.003)	(0.006)		(0.003)		
Female			−0.400***	−0.396***	−0.377***				
			(0.010)	(0.012)	(0.021)				
Years of education			0.020***	0.007	0.014*				
			(0.004)	(0.005)	(0.008)				
Years of education squared			0.004***	0.004***	0.004***				
			(0.000)	(0.000)	(0.000)				
Normalized Raven's				0.071***					
				(0.007)					
Normalized Raven's squared				0.013***					
				(0.005)					
Individual fixed effects	N	N	N	N	N	Y	Y	Y	Y
Time fixed effects	N	Y	Y	Y	Y	Y	Y	Y	Y
Switchers only					Y				
Number of observations	258,745	258,745	258,745	196,354	69,519	258,745	258,745	258,745	258,580
Number of individuals	31,537	31,537	31,537	23,214	5,683	31,537	31,537	31,537	31,530

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TABLE 5. Continued

(B) Kenya

	Dependent variable: log earnings								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8) Log wage	(9) Log real wage
Urban	0.778***	0.660***	0.604***	0.603***	0.297***	0.273***	0.219***	0.123***	0.046
	(0.035)	(0.034)	(0.032)	(0.032)	(0.050)	(0.042)	(0.040)	(0.043)	(0.043)
Log hours		0.348**	0.259*	0.262*	0.309		0.238		
		(0.157)	(0.143)	(0.145)	(0.334)		(0.242)		
Log hours squared		0.009	0.014	0.014	0.013		0.013		
		(0.017)	(0.016)	(0.016)	(0.036)		(0.026)		
Female			−0.455***	−0.431***	−0.299***				
			(0.033)	(0.034)	(0.065)				
Years of education			−0.007	−0.012	−0.050				
			(0.033)	(0.034)	(0.057)				
Years of education squared			0.005***	0.005***	0.006**				
			(0.002)	(0.002)	(0.003)				
Normalized Raven's				0.062***	0.087**				
				(0.020)	(0.038)				
Normalized Raven's squared				−0.027	−0.007				
				(0.018)	(0.032)				
Individual fixed effects	N	N	N	N	N	Y	Y	Y	Y
Time fixed effects	N	Y	Y	Y	Y	Y	Y	Y	Y
Switchers only					Y				
Number of observations	130,322	130,322	130,322	124,481	38,206	130,322	130,322	130,322	130,251
Number of individuals	4,718	4,718	4,718	4,452	1,017	4,718	4,718	4,718	4,715

Notes: Panel A uses data from the IFLS and panel B uses data from the KLPS. Please refer to Section 3 for further details on the data and to the notes of Table 4 for additional information on the variables. For the IFLS, the urban indicator is obtained from survey responses to the question: "Is the area you live in a village, a town or a city?" If the person reports living in a town or city, the urban indicator variable equals 1. For the KLPS, the urban indicator equals 1 if a person lives in a large town or a city. Please see the text in section 3.2 for further details on this classification. Column (5) only includes switchers, who are defined as individuals with at least one observation in both an urban and rural area. Each regression in columns (2)–(8) include quadratic controls for age. All regressions are clustered at the individual level. Robust standard errors are in parentheses.
$*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

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TABLE 6. Gap in earnings, Indonesia for individuals born rural and urban.

(A) Individuals born in rural areas			
	Dependent variable: log earnings		
	(1)	(2)	(3) Log wage
Urban	0.438***	0.210***	0.039**
	(0.014)	(0.011)	(0.017)
Individual fixed effects	N	N	Y
Control variables and time FE	N	Y	Y
Number of observations	186,889	186,889	186,889
Number of individuals	21,764	21,764	21,764
(B) Individuals born in urban areas			
	Dependent variable: log earnings		
	(1)	(2)	(3) Log wage
Urban	0.320***	0.146***	0.013
	(0.027)	(0.021)	(0.027)
Individual fixed effects	N	N	Y
Control variables and time FE	N	Y	Y
Number of observations	71,354	71,354	71,354
Number of individuals	9,662	9,662	9,662

Notes: This table repeats the analyses of Table 5 for those born in rural and urban areas, respectively, in Indonesia. Please refer to Section 3 for further details on the data and to the notes of Table 5 for additional information on the variables. Control variables include log hours worked, log hours worked squared, age, age squared, years of education, years of education squared and an indicator for being female. When also including individual fixed effects in column (3), the control variables are reduced to only log hours worked, log hours worked squared, and age squared because the others are absorbed by the individual fixed effects. When using log wage as the dependent variable in column (3), only age squared is included as a control variable. All regressions are clustered at the individual level. Robust standard errors are in parentheses.
$*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

log wage specification in column (3), gaps are again somewhat larger for those born rural but we cannot reject that the returns are equal (p-value = 0.208). The estimated earnings gain to urban employment is 21 log points for those born in rural areas (panel A) and 15 log points for those born urban (panel B).[28]

28. In Online Appendix B, we explore a richer specification of human capital which permits worker and sector-specific productivity. A comparison of the marginal distributions of productivities in these figures corroborates the importance of absolute advantage in explaining the differences in wages, whereas the observed upward sloping relationship between individual productivity in the two sectors does not suggest that rural workers are uniquely well-suited to urban non-agricultural work.

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There are a number of alternative measures of individual agricultural productivity that are worth considering to assess robustness of the main results. Online Appendix Figure A.3 illustrates how each source of agricultural productivity data in both the IFLS and KLPS contributes to the overall sample, and classifies measures into those that are more reliably measured (e.g., paid employment), and those that are less reliably measured (e.g., measures based on production in subsistence agriculture). We next assess robustness to different definitions of employment in agriculture, including if the majority of hours are in the sector (our main measure), as well as measures that classify an individual as working agriculture is any hours are in the sector, or alternatively if all hours are in the sector. We additionally explore robustness to the use of both labor earnings and self-employed profits in agriculture (main measure), versus measures that use only one or the other. For both Indonesia (Online Appendix Table A.10, panel A) and Kenya (panel B), estimated agricultural productivity gaps remain small and positive across five alternative measures, ranging from −2 to 13 log points in Indonesia and 1 to 10 log points in Kenya.[29]

Online Appendix C describes an analogous analysis using household consumption expenditure data that is considered a useful measure of living standards in development economics, and has the additional advantage of allowing us to include individuals for whom earnings data are not available. Estimated sectoral gaps remain similar using this measure.

Another dimension of welfare relates to patterns of unemployment. Online Appendix Table A.15 explores whether there are differences in unemployment rates and search behavior between urban and rural areas for Kenya, where these data are available. We find that unemployment (measured several ways) is either similar in urban and rural areas (panel A, column (3)) or somewhat higher in urban than in rural areas, conditional on individual fixed effects (panel A, column (6), and panel B), strengthening the main finding that movers to urban areas may not experience large gains in total earnings.[30,31]

29. Further details and robustness checks are contained in Online Appendix Tables A.11–A.14. In Online Appendix Table A.11, we present estimates for Indonesia on a sample of individuals who are at most 30 years old, for greater comparability with the Kenya sample; the estimates remain similar. Online Appendix Tables A.12 and A.13 report results separately for labor earnings and self-employment earnings, respectively, with similar results. Online Appendix Table A.14 reports results for Kenya including subsistence agriculture even when the respondent is not the main decision maker for an activity (panel A), and excluding subsistence agriculture (panel B), and results are robust.

30. In this paper, we consider *mean differences* in productivity or consumption across sectors, but *variability of outcomes* could also be a determinant of individual wellbeing, as well as of migration choices (Munshi and Rosenzweig 2016). We test whether the variability of earnings in the agricultural (rural) sector is different than variability in the non-agricultural (urban) sector, conditional on individual fixed effects, and find mixed results. There are no statistically significant differences in variability across sectors in the Kenya sample. There is significantly more variability in agricultural (rural) wages and earnings in Indonesia relative to the non-agricultural sector (urban areas), although no significant differences in consumption variability (not shown). We leave additional exploration of these issues for future research.

31. Results are unchanged when using alternative approaches to accounting for clustering (Bell and McCaffrey 2002, Cameron and Miller (2015), and Young (2016); see Online Appendix Table A.16.

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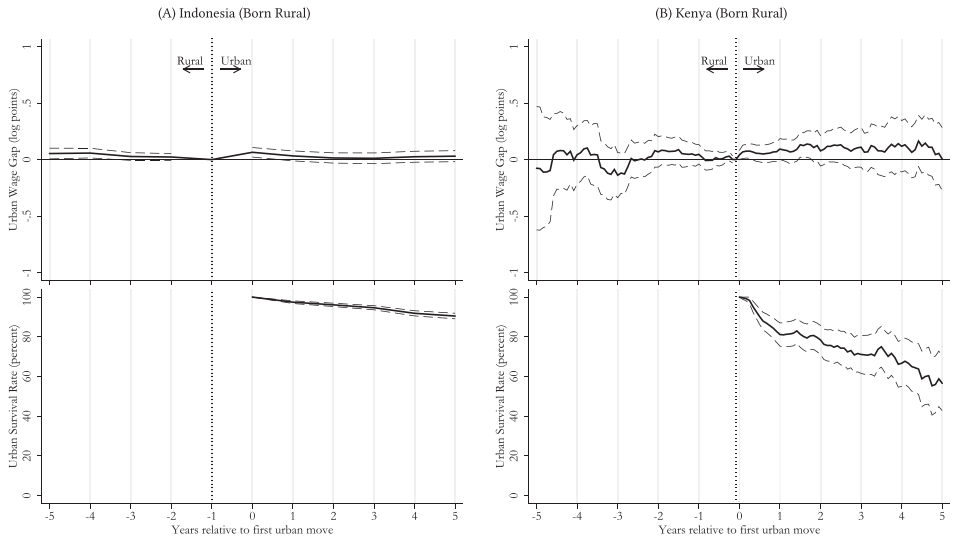


FIGURE 3. Event study of urban migration. panel A uses data from individuals in the IFLS who are born in rural areas, and panel B uses data from the KLPS. Please refer to Section 3 for further details on the data. The top half of each panel reports event study coefficients β_τ from a regression of log wages described in equation (8) (in Section 4.2). The solid line represents the point estimate, and the dashed lines represent the 95% confidence interval. Estimates represent the difference in mean wages between movers and non-movers net of the difference that existed in the period prior to the move. Regressions include individual fixed effects, time fixed effects, squared-age, and indicator variables that pool observations exceeding a 5-year window of the move. The lower half of each panel reports the fraction of people having no rural observations from period zero to the period of interest. (The estimated fraction of survivors can in principle increase due to sample composition changes as can be seen in the lower half of panel B.) In the IFLS, there are 2,247 individuals who have observed wages in the year of the move and the year prior; 1,677 of these individuals report wages 5 years later. In the KLPS, these numbers are 336 and 53, respectively.

4.2. Dynamics of the Productivity Gap

In unpacking the main result, we examine if dynamics and experience effects produce productivity gains that do not materialize right away. In particular, the main specifications capture the average differences in earnings and wages between rural and urban areas, but the presence of experience effects may cause individuals to earn more after spending time in urban areas. Figure 3 presents event study analyses of whether individuals earn more per hour worked (w_{it}) after migrating, where we estimate regressions of the form:

$$w_{it} = \theta_i + \delta_t + \boldsymbol{X}'_{it}\boldsymbol{b} + \sum_\tau \beta_\tau 1(\text{Urban})_{i,t+\tau} + \varepsilon_{it} \tag{8}$$

These regressions are estimated on an unbalanced panel of individual-time periods and include individual fixed effects θ_i, time fixed effects δ_t, and controls X_{it} for squared age. The Indonesian and Kenyan analyses both condition on individuals being

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born rural. The specifications include event study indicators that cover the 5 years preceding and succeeding the move. The corresponding coefficients β_τ are the primary parameters of interest. We exclude an indicator for $\tau = -1$, and thus the estimates are relative to the period immediately prior to the move. We require that individuals in our sample have a wage observations in period $\tau = -1$ and in the period of the move $\tau = 0$, but do not enforce a requirement that individuals have a wage measurement for every single period in the 5 years before and after the move.[32]

Because the composition of workers in the labor force may change, we interpret the results with caution, but nonetheless believe that the novelty of a rich panel dataset is worth exploring. For instance, the exercise allows us to also assess wage dynamics *prior* to the move. If rural individuals experience a negative wage realization prior to the move, then we should see negative pre-trends, akin to an Ashenfelter (1978) wage dip. Correspondingly, if migrants learn about better urban jobs over time, one should see wages grow as time passes after the move.

In Indonesia, there is some evidence of a slight urban wage gain in the year of the move, but limited evidence of a gain up to 5 years after the urban move (Figure 3, panel A). There are broadly similar results in Kenya relative to the month prior to the move; there is some suggestive indication of slightly rising wages in the first 2 years of urban residence, but these are small (panel B). There is little indication of meaningful premove trends in either country.

In this analysis, we consider wages for individuals who made an urban move regardless of whether they remained in cities or towns, or later moved back to rural areas. The bottom half of panel A shows a "survival" rate in urban areas of over 85% after 5 years in Indonesia, but a rate of less than 60% after 5 years in Kenya, suggesting substantial return migration in the latter. Naturally, one might suspect that those with the worst economic outcomes in urban areas might return home, yet this does not appear to be the case: Online Appendix Figure A.4 separately plots postmove wages for those who remain in urban areas and those who return to rural areas, and we find no evidence of a significant divergence in earnings between these two groups (although note that this analysis has limited statistical power). This suggests a direction for future research in uncovering the reasons for these moves, including the role played by noneconomic factors, including family reasons and heterogeneity in the taste for urban living.[33]

Other scholars have argued that job experience is particularly valuable in big cities and that residence in these cities may boost individual productivity over time (see de la Roca and Puga 2017 for the Spanish case). We examine this issue in Online Appendix D, estimating productivity gaps for the largest five cities in Indonesia and Kenya. Although some cities show modestly larger productivity gaps, including Bekasi

32. The vector of control variables also includes indicators for individual-time periods outside of our main ±5-year window of analysis (on both ends)

33. We carry out an analogous event study of moves to rural areas among those born urban in Indonesia, and similarly find no evidence of significant dynamic impacts (see Online Appendix Figures A.5 and A.6).

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in Indonesia and Nairobi and Mombasa in Kenya, there is not a consistent pattern across big cities.

5. Conclusion

Several influential recent studies document large sectoral productivity gaps in low-income countries and highlight an apparent puzzle, namely, "why so many workers remain in the agricultural sector, given the large residual productivity gaps with the rest of the economy" (Gollin, Lagakos, and Waugh 2014, p. 941). This study makes two main contributions using data from low-income countries with large populations (Indonesia and Kenya) located in two different regions. First, we show that estimating sectoral productivity gaps—both across non-agricultural and agricultural sectors, and across urban and rural areas—using panel data and including individual fixed effects leads to a reduction of roughly 67%–92% in the estimated gaps. The second main empirical contribution lies in demonstrating that there is extensive individual selection across sectors, both along typically observable dimensions such as education as well as measures of skill (here, cognitive ability) that most standard economic datasets lack.

Taken together, the findings point to the role that individual selection plays in driving observed sectoral gaps and could call into question strong causal interpretations. As a result, the puzzle of why the share of workers in rural agriculture remains high may not be as much of a puzzle as previously thought. Similarly, if observed gaps are partly driven by selection, then policies to incentivize workers to move to urban areas (and out of agriculture), based on the logic of input misallocation, might not appreciably raise aggregate living standards and would be a less attractive policy direction.

Our main productivity gap estimates are derived from individual movers, namely, those with productivity measured in both sectors. Thus, as discussed previously, a logical way to reconcile our finding of small sectoral gaps with the existing macroeconomic empirical evidence of large average gaps is the possibility that productivity effects among non-movers would be much larger than those of movers. Given the nature of our data, we acknowledge that it is impossible to rule out this possibility without imposing more structure, and it clearly merits further investigation, although the lack of measured individual productivity in both sectors for non-movers naturally complicates the rigorous identification of these relationships.

Several factors lean against this interpretation in our view, at least in the short run. First, it is natural to think of the migration decision in terms of a Roy (1951) model, as we do here, in which those with the largest net utility benefits are most likely to move. Under some conditions, this could lead our estimates to overstate gaps between sectors overall. Although it is possible that those individuals who remain in the rural agricultural sector might receive the largest positive earnings gains from moving, their choice not to do so might simply reflect high financial or nonfinancial costs to migration. For instance, the amenities found in a large city are quite different than those in rural areas, and individuals may have strong and heterogeneous preferences for them, leading to large reductions in utility for some migrants even if wages rise. Poor

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individuals may also face credit constraints or other financial frictions that prevent them from exploiting wage gaps, and easing these constraints could boost migration rates, as argued for Indonesia by Kleemans (2016) and Bazzi (2017) and India by Munshi and Rosenzweig (2016). We are unable to rule out these possibilities. However, the long timeframes of both datasets used in this study help to, at least partially, mitigate this concern: Some poor individuals with high returns to migration presumably had access to improved credit at *some* point during 1988–2015 in Indonesia or 1998–2014 in Kenya and managed to move.

A promising approach to estimating the returns to migration in low-income countries among those who are typically "non-movers" and may face such constraints is the Bryan, Chowdhury, and Mobarak (2014) study in Bangladesh. They find that a modest subsidy induces 22% of households to send a migrant to towns and cities for temporary work during the agricultural low season; the relatively low rate of migration may indicate that the utility costs of migration are nontrivial. Among movers, there is an estimated increase in per capita consumption among the sending household of roughly 30% over 2 years and 25% average gain in earnings (not statistically significant) among those assigned to the subsidy. Overall, the study provides some indication of positive returns to temporary seasonal migration. Nonetheless, the earnings gains are fairly modest in size and are closer in magnitude to the small gaps we estimate in this paper than to those found in many other recent contributions. It is also worth noting that the subsidy was delivered during times of the year (the agricultural low season) when agricultural productivity was thought to be particularly meager and targeted to regions thought especially likely to benefit from seasonal moves. The consumption treatment effects measured in Bryan et al. (2014) might be smaller had they been measured at other times of the year or targeted to different regions.

The case of urban-born non-movers is less well understood. Recall (from Table 2) that individuals raised in urban areas have higher cognitive scores (on a test of fluid intelligence) than those raised in rural areas. This gap raises two intriguing possibilities. The first is that wave after wave of rural to urban (urban to rural) migration by positively (negatively) selected individuals over many decades, combined with partial heritability of cognitive ability, have reshaped the underlying ability distributions in these two sectors. This would simply be an intergenerational extension of the patterns of individual selection across urban and rural areas that we and Young (2013) document, and would not necessarily change the interpretation of our main results.

Another explanation, which is not mutually exclusive, is that there is a lower cost to skill acquisition in urban areas, either due to improved provision of schooling there or something about the nature of social interactions (e.g., the density of interactions or other forms of childhood intellectual stimulation). In other words, given the importance of early childhood circumstances for lifetime cognitive development (e.g., Gertler et al. 2014), growing up in a city might generate higher average adult skill levels. This would generate a positive causal effect of urban residence on labor productivity, albeit in the very long run and on the movers' children; anecdotally, many migrants do claim to move in order to improve their children's wellbeing more than their own. These effects would not be captured in the 5-year follow-up period that we consider in this study

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(in Figure 3), but could be contributing to large, persistent and real causal urban-rural productivity gaps overall. Indeed, Nakamura, Sigurdsson, and Steinsson (2020) study migration induced by a volcanic explosion in Iceland, and show that adult movers gain little from moving out of a rural area, but their children earn far more.

We tentatively assess this possibility using the IFLS data and a separate dataset collected among young children of KLPS respondents, and find suggestive evidence that urban residence may have positive impacts on the cognitive ability of the next generation. In particular, young children born in urban areas have significantly higher cognitive scores than rural-born children, even conditional on the parent's own schooling and fluid intelligence (Online Appendix Table A.17 and Online Appendix Figure A.7). The conditional average gaps are approximately 0.03 standard deviation units in Indonesia (Online Appendix Table A.17, panel A, column (5)) and 0.26 in Kenya (panel B, column (5)), although we cannot rule out the existence of some omitted variable bias.

Some natural next steps for research include: extending long-run panel data analysis to new settings (as data become available); conducting more experiments that induce migration, generating "local" estimates in new sub-populations and improving understanding of migration constraints; and exploring the intergenerational effects of sectoral and residential choice.

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Supplementary Data

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